COS 433: Cryptography

Announcements:

- Problem Set 4 due Wed 11/12

Digital signatures

One-Time Signatures

$\mathrm{Gen}(1^\lambda) \rightarrow (\mathrm{sk}, \mathrm{vk})$

$m \in \{0,1\}^n$

vk (broadcast)

$\longrightarrow$

$(m, \sigma = \mathrm{Sign}(\mathrm{sk}, m))$

$\longrightarrow$

$\mathrm{Verify}(\mathrm{vk}, m, \sigma) \in \{0, 1\}$

Theorem: Assuming one-way functions exist, for any $n(\lambda) = \mathbf{poly}(\lambda)$, there exist one-time signatures with message length $n(\lambda)$!

- Key length: $|\mathrm{vk}| = n(\lambda) \cdot \lambda > n(\lambda)$ (long keys)
- We will fix this ~~later~~ now

One-time signatures with short keys

One-Time Signatures

$\mathrm{Gen}(1^{\lambda}) \rightarrow (\mathrm{sk}, \mathrm{vk})$

$m \in \{0,1\}^n$

vk (broadcast)

$(m, \sigma = \mathrm{Sign}(\mathrm{sk}, m))$

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- Suppose we start with a scheme with large vk
- Do PRGs help?
  - Can compress sk but not vk
- Not clear how to compress vk.

Hash-and-Sign Paradigm

$\mathrm{Gen}(1^{\lambda}) \rightarrow (\mathrm{sk}, \mathrm{vk})$

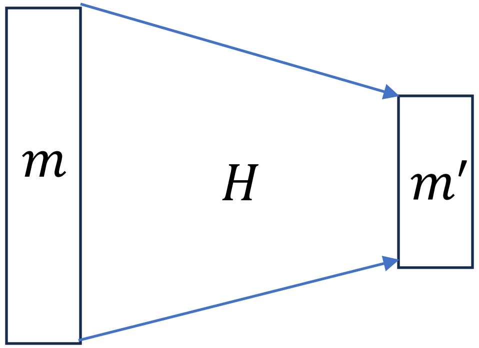
$m \in \{0,1\}^n$

vk (broadcast)

$(m, \sigma = \mathrm{Sign}(\mathrm{sk}, m))$

$\mathrm{Verify}(\mathrm{vk}, m, \sigma) \in \{0, 1\}$

- Idea: extend the scheme to handle longer messages with (roughly) same vk

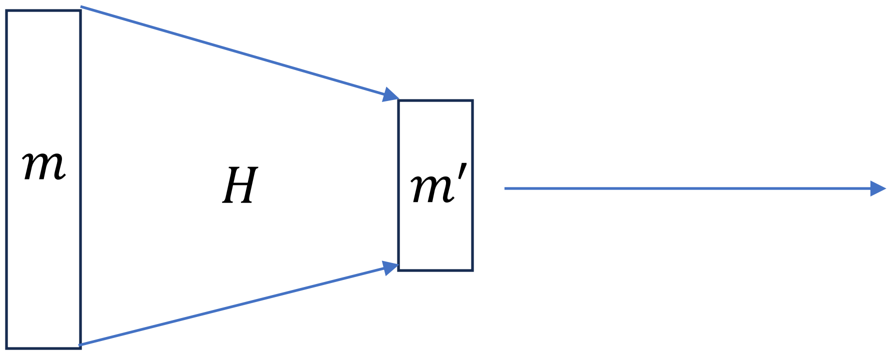


$\sigma = \mathrm{Sign}(\mathrm{sk}, H(m))$

- H must be a publicly computable ”hash function” (so that σ can be verified)

so (s

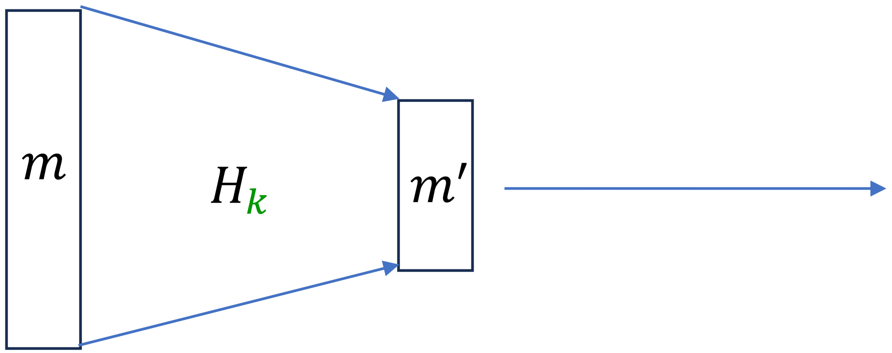
Hash-and-Sign Paradigm



$\sigma = \mathrm{Sign}(\mathrm{sk}, H(m))$

- Is this secure?
- If H is a fixed function: no!
  - Suppose that $H(m_1) = H(m_2)$. Then can forge on m_2 given signature on m_1.

Hash-and-Sign Paradigm



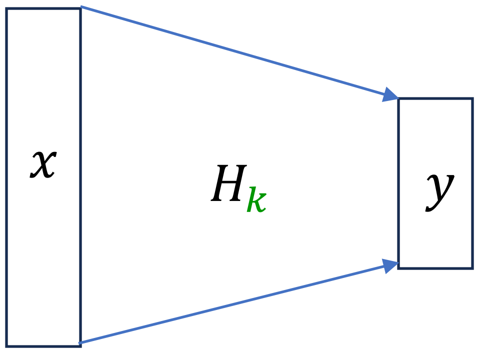
$$\sigma = \mathrm{Sign}(\mathrm{sk}, H_k(m))$$

- Is this secure?

- If H is a fixed function: no!
  - Suppose that $H(m_1) = H(m_2)$. Then can forge on m_2 given signature on m_1.

- Instead: use a keyed hash function H_k
  - Add k to both the signing and verification keys

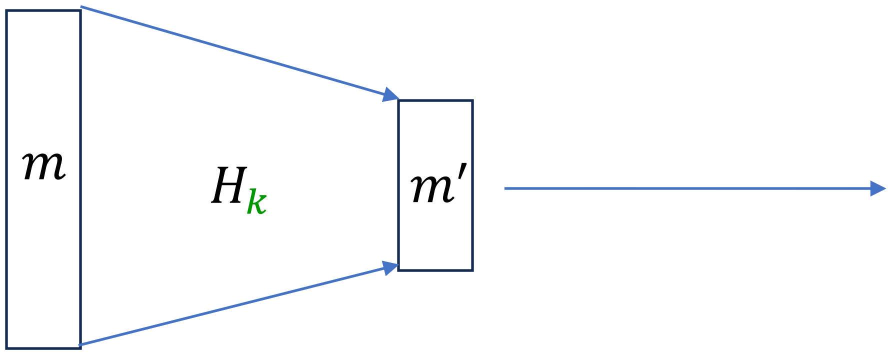
Collision-Resistant Hash Functions



Definition: A family of efficiently computable functions $\mathcal{H} = \left\{H_k : \{0,1\}^{n(\lambda)} \to \{0,1\}^{\lambda}\right\}_{k \in K}$ is a collision-resistant hash function (CRHF) family with input length $n = n(\lambda)$ if for all polynomial-time A,

$$\Pr[A(k) \to (x_1, x_2) : x_1 \neq x_2 \text{ and } H_k(x_1) = H_k(x_2)] = \mathrm{negl}(\lambda)$$

Collision-Resistant Hash Functions



$\sigma = \mathrm{Sign}(\mathrm{sk}, H_k(m))$

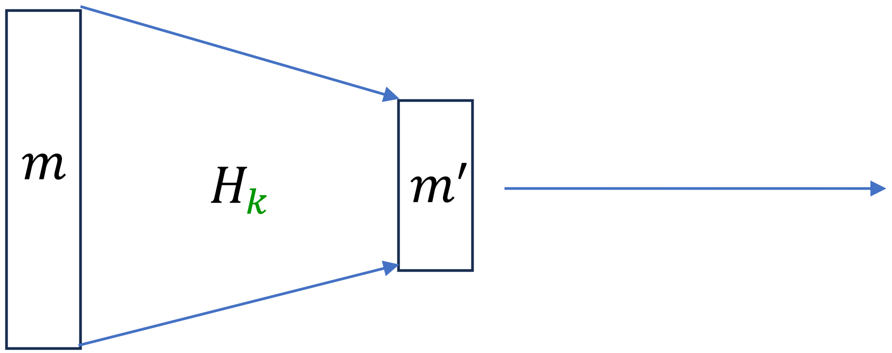
Theorem: If

- $(\mathrm{Gen}, \mathrm{Sign}, \mathrm{Verify})$ is a (one time OR full) digital signature scheme with message length $n = n(\lambda)$, and
- $\mathcal{H}$ is a collision-resistant hash function (CRHF) family with output length n,

then $(\mathrm{Sign} \circ \mathcal{H}, \mathrm{Verify} \circ \mathcal{H})$ is also a (one time OR full) digital signature scheme with

$$\mathrm{Gen}'(1^\lambda) \to ((\mathrm{sk}, k), (\mathrm{vk}, k))$$

Collision-Resistant Hash Functions



$\sigma = \mathrm{Sign}(\mathrm{sk}, H_k(m))$

So if we can write down a CHRF family with short hash keys, we can sign (single) long messages with short $(\mathrm{sk}, \mathrm{vk})$!

Intuition: $H_k = \mathrm{PRF}_k$ would work… if k were secret.

Two kinds of constructions

- Algebraic CRHFs based on, e.g., discrete log
- "Obscure programs" approach with a heuristic analysis: random oracle model

CRHFs based on discrete log

$H_k\colon \mathbb{Z}_q \times \mathbb{Z}_q \to \mathbb{Z}_p$

$p = 2q + 1$

$k = (g, h)$

g generator for $\mathbf{QR}_p \subset \mathbb{Z}_p^\times$

$h \leftarrow \mathbf{QR}_p$

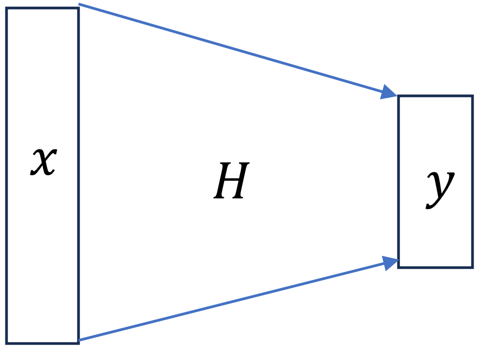
$$H_k(x, y) = g^x h^y$$

CRHF Length Extension: Merkle-Damgard

(on board)

Random Oracle Heuristic

1. Design a crypto scheme that makes black-box use of a hash function H.



2. Prove security "in the random oracle model:"
   - H is a uniformly random function
   - Adversary is of the form $A^{H(\cdot)}$, making black-box calls to H.

3. Heuristically instantiate with a concrete hash function family $\{H_k\}$

4. Philosophy (unproven): carefully designed hash functions should not have non-black-box attacks.

Random Oracle Heuristic

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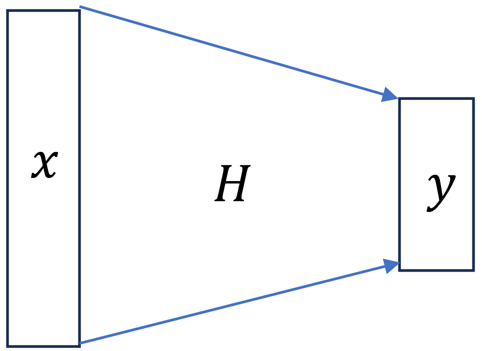
4. Example: H is a CRHF in the random oracle model (either unkeyed or salted: $H_k(x) = H(k, x)$)



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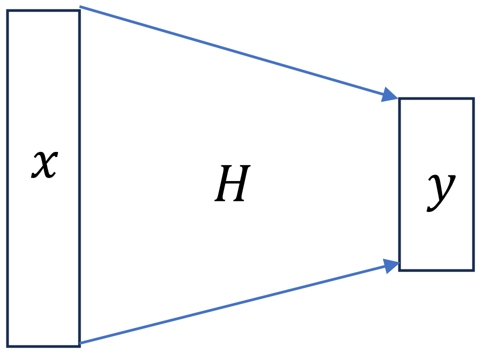

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Extremely common/powerful tool for designing practical cryptography

But: sometimes unsound, need to be careful continued next time